

BRAC University

Department of Mathematics and Physical Sciences

MAT110

Lecture 13

**Gradient, Lagrange Multipliers,
Applications of Lagrange Multipliers
(Optimization)**

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Contents

1	The Gradient	1
2	Lagrange Multipliers	1
2.1	Example 1	2
2.2	Example 2	4
3	Optimization Problem Using Lagrange Multipliers	7
4	Problem Set	11

1. The Gradient

The Gradient

(a) If f is a function of x and y , then the gradient of f is defined by

$$\nabla f(x, y) = f_x(x, y)\hat{\mathbf{i}} + f_y(x, y)\hat{\mathbf{j}}.$$

(b) If $f = f(x, y, z)$, then

$$\nabla f(x, y, z) = f_x(x, y, z)\hat{\mathbf{i}} + f_y(x, y, z)\hat{\mathbf{j}} + f_z(x, y, z)\hat{\mathbf{k}}.$$

One of the mathematical meanings of the gradient is that the gradient vector determines the orientation of the tangent plane at a particular point.

2. Lagrange Multipliers

Theorem 13.9.3: Constrained-Extremum Principle for Two Variables and One Constraint

Let f and g be functions of two variables with continuous first partial derivatives on some open set containing the constraint $g(x, y) = 0$, and assume that $\nabla g \neq \mathbf{0}$ at any point on this curve.

If f has a constrained relative extremum, then this extremum occurs at a point (x_0, y_0) on the constraint curve at which the gradient vectors $\nabla f(x_0, y_0)$ and $\nabla g(x_0, y_0)$ are parallel; that is, there is some number λ such that

$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0). \quad (1)$$

This scalar λ is called a **Lagrange multiplier**.

Thus, the method of Lagrange multipliers for finding constrained relative extrema is to look for points on the constraint curve $g(x, y) = 0$ at which equation (1) is satisfied for some scalar λ .

2.1. Example 1

Example 01

At what point or points on the circle $x^2 + y^2 = 1$ does $f(x, y) = xy$ have an absolute maximum, and what is that maximum?

Answer

The circle $x^2 + y^2 = 1$ is a closed and bounded set, and $f(x, y) = xy$ is a continuous function. Therefore, according to the Extreme-Value Theorem, $f(x, y)$ has an absolute maximum and an absolute minimum on the circle $x^2 + y^2 = 1$.

To find these extrema, we use Lagrange multipliers to find the constrained relative extrema, and then evaluate f at those relative extrema to find the absolute extrema. At first, we want to maximize

$$f(x, y) = xy$$

subject to the constraint

$$g(x, y) = x^2 + y^2 - 1 = 0. \quad (2)$$

According to Theorem 13.9.3,

$$\nabla f(x, y) = \lambda \nabla g(x, y), \quad \lambda \in \mathbb{R}. \quad (3)$$

Here,

$$\begin{aligned} \nabla f(x, y) &= \left(\frac{\partial}{\partial x} \hat{\mathbf{i}} + \frac{\partial}{\partial y} \hat{\mathbf{j}} \right) f(x, y) \\ &= \frac{\partial f}{\partial x} \hat{\mathbf{i}} + \frac{\partial f}{\partial y} \hat{\mathbf{j}} \\ &= \frac{\partial}{\partial x} (xy) \hat{\mathbf{i}} + \frac{\partial}{\partial y} (xy) \hat{\mathbf{j}} \\ &= y \hat{\mathbf{i}} + x \hat{\mathbf{j}}. \end{aligned} \quad (4)$$

Also,

$$\begin{aligned} \nabla g(x, y) &= \frac{\partial g}{\partial x} \hat{\mathbf{i}} + \frac{\partial g}{\partial y} \hat{\mathbf{j}} \\ &= \frac{\partial}{\partial x} (x^2 + y^2 - 1) \hat{\mathbf{i}} + \frac{\partial}{\partial y} (x^2 + y^2 - 1) \hat{\mathbf{j}} \\ &= 2x \hat{\mathbf{i}} + 2y \hat{\mathbf{j}}. \end{aligned} \quad (5)$$

Now,

$$\nabla g(x, y) = \mathbf{0} \iff 2x \hat{\mathbf{i}} + 2y \hat{\mathbf{j}} = 0 \hat{\mathbf{i}} + 0 \hat{\mathbf{j}} \iff x = 0, y = 0.$$

The gradient of $g(x, y)$ is zero only at $(0, 0)$, but $(0, 0)$ does not belong to the given circle $x^2 + y^2 - 1 = 0$. Hence,

$$\nabla g \neq \mathbf{0} \quad \text{at every point on the circle } x^2 + y^2 = 1.$$

From (3),

$$\begin{aligned}\nabla f &= \lambda \nabla g, \\ y\hat{\mathbf{i}} + x\hat{\mathbf{j}} &= \lambda(2x\hat{\mathbf{i}} + 2y\hat{\mathbf{j}}), \\ y\hat{\mathbf{i}} + x\hat{\mathbf{j}} &= 2x\lambda\hat{\mathbf{i}} + 2y\lambda\hat{\mathbf{j}}.\end{aligned}$$

Equating components gives

$$y = 2x\lambda, \quad x = 2y\lambda.$$

Thus,

$$\lambda = \frac{y}{2x} = \frac{x}{2y},$$

and hence

$$y^2 = x^2. \tag{6}$$

Putting $y^2 = x^2$ into the constraint (2),

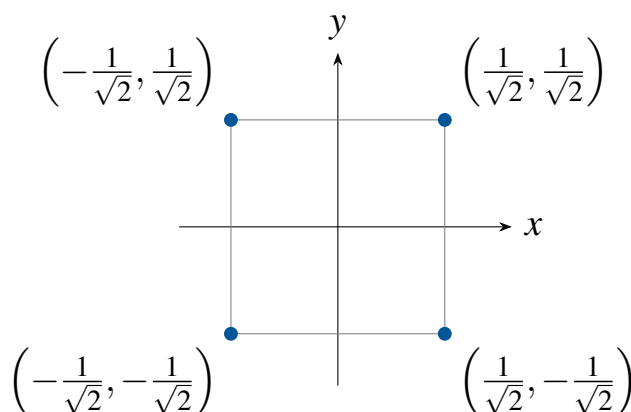
$$\begin{aligned}x^2 + y^2 - 1 &= 0, \\ x^2 + x^2 - 1 &= 0, \\ 2x^2 &= 1, \\ x &= \pm \frac{1}{\sqrt{2}}.\end{aligned}$$

Also,

$$y^2 = x^2 = \frac{1}{2} \implies y = \pm \frac{1}{\sqrt{2}}.$$

Therefore, the relative extrema occur at the points

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \quad \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \quad \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \quad \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right).$$



(x, y)	$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$	$\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$	$\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$	$\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$
$f(x, y)$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$

Thus, the function $f(x, y) = xy$ has an absolute maximum of $\frac{1}{2}$, occurring at the two points

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \quad \text{and} \quad \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right).$$

Also, f has an absolute minimum of $-\frac{1}{2}$, occurring at the points

$$\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) \quad \text{and} \quad \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right).$$

2.2. Example 2

Example 02

Use Lagrange multipliers to find the maximum and minimum values of $f(x, y, z) = xyz$ subject to the constraint

$$x^2 + y^2 + z^2 = 1.$$

Also find the points at which these extreme values occur.

Answer

Given

$$f(x, y, z) = xyz, \tag{1}$$

and the constraint

$$g(x, y, z) = x^2 + y^2 + z^2 - 1 = 0. \tag{2}$$

The function $f(x, y, z) = xyz$ is continuous in $\mathbb{R}^3$, and the constraint $g(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$ is a sphere of unit radius centred at $(0, 0, 0)$ that is closed and bounded. Therefore, according to the Extreme-Value Theorem, the function has an absolute maximum and an absolute minimum on $g(x, y, z) = 0$.

Now,

$$\begin{aligned} \nabla f &= \frac{\partial f}{\partial x} \hat{\mathbf{i}} + \frac{\partial f}{\partial y} \hat{\mathbf{j}} + \frac{\partial f}{\partial z} \hat{\mathbf{k}} \\ &= yz \hat{\mathbf{i}} + xz \hat{\mathbf{j}} + xy \hat{\mathbf{k}}, \end{aligned} \tag{3}$$

$$\nabla g = \frac{\partial g}{\partial x} \hat{\mathbf{i}} + \frac{\partial g}{\partial y} \hat{\mathbf{j}} + \frac{\partial g}{\partial z} \hat{\mathbf{k}}$$

$$= 2x\hat{i} + 2y\hat{j} + 2z\hat{k}. \quad (4)$$

According to Theorem 13.9.3,

$$\nabla f = \lambda \nabla g, \quad \lambda \in \mathbb{R}.$$

Therefore,

$$\begin{aligned} yz\hat{i} + xz\hat{j} + xy\hat{k} &= \lambda(2x\hat{i} + 2y\hat{j} + 2z\hat{k}), \\ yz &= 2x\lambda, \\ xz &= 2y\lambda, \\ xy &= 2z\lambda. \end{aligned}$$

Hence,

$$\frac{yz}{2x} = \frac{xz}{2y} = \frac{xy}{2z}. \quad (5)$$

From (5),

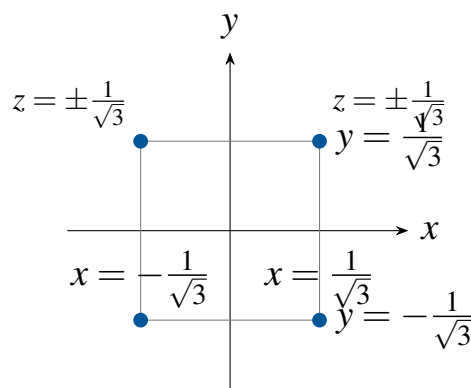
$$y^2 = x^2, \quad z^2 = x^2.$$

Putting $y^2 = x^2$ and $z^2 = x^2$ into equation (2),

$$\begin{aligned} x^2 + x^2 + x^2 &= 1, \\ 3x^2 &= 1, \\ x &= \pm \frac{1}{\sqrt{3}}. \end{aligned}$$

Also,

$$y = \pm \frac{1}{\sqrt{3}}, \quad z = \pm \frac{1}{\sqrt{3}}.$$



There are eight possibilities. Evaluating $f(x, y, z) = xyz$ gives:

No.	Point	$f(x, y, z) = xyz$	Comment
1	$\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$	$\frac{1}{3\sqrt{3}}$	Max
2	$\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$	$-\frac{1}{3\sqrt{3}}$	Min
3	$\left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$	$-\frac{1}{3\sqrt{3}}$	Min
4	$\left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$	$\frac{1}{3\sqrt{3}}$	Max
5	$\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$	$-\frac{1}{3\sqrt{3}}$	Min
6	$\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$	$\frac{1}{3\sqrt{3}}$	Max
7	$\left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$	$\frac{1}{3\sqrt{3}}$	Max
8	$\left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$	$-\frac{1}{3\sqrt{3}}$	Min

Thus, the maximum value is

$$\boxed{\frac{1}{3\sqrt{3}}},$$

which occurs at the points

$$\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \quad \left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \\ \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right), \quad \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right).$$

The minimum value is

$$\boxed{-\frac{1}{3\sqrt{3}}},$$

which occurs at the points

$$\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \quad \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \\ \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right), \quad \left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right).$$

3. Optimization Problem Using Lagrange Multipliers

Example 03: Optimization Problem

Use the method of Lagrange multipliers to find the dimensions of a rectangle with perimeter p and maximum area.

Answer

Let

x = length of the rectangle, y = width of the rectangle, A = area of the rectangle.

We want to maximize

$$A = xy$$

on the line segment

$$2x + 2y = p, \quad 0 \leq x, y, \quad (7)$$

which corresponds to the perimeter constraint. This line segment is a closed and bounded set, and since $f(x, y) = xy$ is a continuous function, it follows from the Extreme-Value Theorem (Theorem 13.8.3) that f has an absolute maximum on this segment. This absolute maximum must also be a constrained relative maximum since f is 0 at the endpoints of the segment and positive elsewhere on the segment. If

$$f(x, y) = xy \quad \text{and} \quad g(x, y) = 2x + 2y,$$

then

$$\nabla f = y\hat{\mathbf{i}} + x\hat{\mathbf{j}} \quad \text{and} \quad \nabla g = 2\hat{\mathbf{i}} + 2\hat{\mathbf{j}}.$$

Noting that $\nabla g \neq \mathbf{0}$, the Lagrange multiplier condition gives

$$\nabla f = \lambda \nabla g.$$

Therefore, at a constrained relative maximum,

$$y\hat{\mathbf{i}} + x\hat{\mathbf{j}} = \lambda (2\hat{\mathbf{i}} + 2\hat{\mathbf{j}}).$$

This is equivalent to the two equations

$$y = 2\lambda \quad \text{and} \quad x = 2\lambda.$$

Eliminating λ from these equations, we obtain

$$x = y,$$

which shows that the rectangle is actually a square. Using this condition and the constraint (7), we obtain

$$2x + 2y = p,$$

$$\begin{aligned}2x + 2x &= p, \\4x &= p, \\x &= \frac{p}{4}.\end{aligned}$$

Since $x = y$,

$$y = \frac{p}{4}.$$

Thus, the rectangle with perimeter p and maximum area is a square with dimensions

$$x = \frac{p}{4}, \quad y = \frac{p}{4}.$$

Example 04: Optimization Problem

Find the points on the sphere

$$x^2 + y^2 + z^2 = 36$$

that are closest to and farthest from the point $(1, 2, 2)$.

Answer

To avoid radicals, we find the points on the sphere that minimize and maximize the *square* of the distance to $(1, 2, 2)$. Thus, we want to find the relative extrema of

$$f(x, y, z) = (x - 1)^2 + (y - 2)^2 + (z - 2)^2$$

subject to the constraint

$$x^2 + y^2 + z^2 = 36. \tag{8}$$

If we let

$$g(x, y, z) = x^2 + y^2 + z^2,$$

then

$$\nabla g = 2x\hat{\mathbf{i}} + 2y\hat{\mathbf{j}} + 2z\hat{\mathbf{k}}.$$

Thus, $\nabla g = \mathbf{0}$ if and only if $x = y = z = 0$. It follows that $\nabla g \neq \mathbf{0}$ at every point of the sphere (8), and hence the constrained relative extrema must occur at points where

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z).$$

That is,

$$2(x - 1)\hat{\mathbf{i}} + 2(y - 2)\hat{\mathbf{j}} + 2(z - 2)\hat{\mathbf{k}} = \lambda (2x\hat{\mathbf{i}} + 2y\hat{\mathbf{j}} + 2z\hat{\mathbf{k}}),$$

which leads to the equations

$$2(x - 1) = 2x\lambda, \quad 2(y - 2) = 2y\lambda, \quad 2(z - 2) = 2z\lambda. \tag{9}$$

We may assume that x , y , and z are nonzero, since $x = 0$ does not satisfy the first equation, $y = 0$ does not satisfy the second equation, and $z = 0$ does not satisfy the third equation. Thus, equation (9) can be rewritten as

$$\frac{x-1}{x} = \lambda, \quad \frac{y-2}{y} = \lambda, \quad \frac{z-2}{z} = \lambda.$$

The first two equations imply that

$$\frac{x-1}{x} = \frac{y-2}{y},$$

from which it follows that

$$y = 2x. \quad (10)$$

Similarly, the first and third equations imply that

$$z = 2x. \quad (11)$$

Substituting (10) and (11) into the constraint equation (8), we obtain

$$\begin{aligned} x^2 + (2x)^2 + (2x)^2 &= 36, \\ 9x^2 &= 36, \\ x &= \pm 2. \end{aligned}$$

Substituting these values into (10) and (11) yields the two points

$$(2, 4, 4) \quad \text{and} \quad (-2, -4, -4).$$

Now,

$$f(2, 4, 4) = 9 \quad \text{and} \quad f(-2, -4, -4) = 81.$$

Therefore,

$$(2, 4, 4) \text{ is the closest point}$$

and

$$(-2, -4, -4) \text{ is the farthest point}.$$

Example 05: Optimization Problem

Use Lagrange multipliers to determine the dimensions of a rectangular box, open at the top, having a volume of 32 ft^3 , and requiring the least amount of material for its construction.

Answer

Let x and y be the dimensions of the base and let z be the height. Since the box is open at the top, the problem is to minimize the surface area

$$S = xy + 2xz + 2yz$$

subject to the volume constraint

$$xyz = 32. \quad (12)$$

If we let

$$f(x, y, z) = xy + 2xz + 2yz \quad \text{and} \quad g(x, y, z) = xyz,$$

then

$$\begin{aligned} \nabla f &= (y + 2z)\hat{\mathbf{i}} + (x + 2z)\hat{\mathbf{j}} + (2x + 2y)\hat{\mathbf{k}}, \\ \nabla g &= yz\hat{\mathbf{i}} + xz\hat{\mathbf{j}} + xy\hat{\mathbf{k}}. \end{aligned}$$

It follows that $\nabla g \neq \mathbf{0}$ at any point on the surface $xyz = 32$, since x , y , and z are all nonzero on this surface. Thus, at a constrained relative extremum we must have $\nabla f = \lambda \nabla g$; that is,

$$(y + 2z)\hat{\mathbf{i}} + (x + 2z)\hat{\mathbf{j}} + (2x + 2y)\hat{\mathbf{k}} = \lambda (yz\hat{\mathbf{i}} + xz\hat{\mathbf{j}} + xy\hat{\mathbf{k}}).$$

This condition yields the three equations

$$y + 2z = \lambda yz, \quad x + 2z = \lambda xz, \quad 2x + 2y = \lambda xy.$$

Because x , y , and z are nonzero, these equations can be rewritten as

$$\frac{1}{z} + \frac{2}{y} = \lambda, \quad \frac{1}{z} + \frac{2}{x} = \lambda, \quad \frac{2}{y} + \frac{2}{x} = \lambda.$$

From the first two equations,

$$y = x. \quad (13)$$

From the first and third equations,

$$z = \frac{1}{2}x. \quad (14)$$

Substituting (13) and (14) into the volume constraint (12) yields

$$\frac{1}{2}x^3 = 32.$$

This equation, together with (13) and (14), yields

$$x = 4, \quad y = 4, \quad z = 2.$$

Therefore, the open-top rectangular box requiring the least amount of material has dimensions

$$\boxed{4 \text{ ft} \times 4 \text{ ft} \times 2 \text{ ft}},$$

where the base is $4 \text{ ft} \times 4 \text{ ft}$ and the height is 2 ft .

4. Problem Set

Use Lagrange multipliers to find the maximum and minimum values of f subject to the given constraint. Also, find the points at which these extreme values occur.

1. $f(x, y) = xy; \quad 4x^2 + 8y^2 = 16$

2. $f(x, y) = x^2 - y^2; \quad x^2 + y^2 = 25$

3. $f(x, y) = 4x^3 + y^2; \quad 2x^2 + y^2 = 1$

4. $f(x, y) = x - 3y - 1; \quad x^2 + 3y^2 = 16$

5. $f(x, y, z) = 2x + y - 2z; \quad x^2 + y^2 + z^2 = 4$

6. $f(x, y, z) = 3x + 6y + 2z; \quad 2x^2 + 4y^2 + z^2 = 70$

7. $f(x, y, z) = xyz; \quad x^2 + y^2 + z^2 = 1$

8. $f(x, y, z) = x^4 + y^4 + z^4; \quad x^2 + y^2 + z^2 = 1$